

Empowering Medical Record Management: Development of a Patient-Centric AI System for Enhanced Medical Data Efficiency

Research Hypothesis and Objectives

Primary hypothesis: The development and implementation of a patient-centric AI system for enhanced electronic health record integration will significantly reduce documentation times and improve workflow efficiency within the NHS.

This hypothesis is rooted in the recognition of the escalating administrative burden faced by healthcare professionals and the pressing need for innovative solutions to alleviate strain on an already stretched workforce. The introduction of GPT technology, renowned for its capability to understand and generate human-like conversations, plays a significant role in achieving the claimed hypothesis. The transformative potential of this research lies in its capacity to revolutionise healthcare documentation practices, moving towards more efficient and accurate record-keeping processes. Our research is novel because no other healthcare system in the world has yet implemented GPT and AI technology to this capacity before. Additionally, the research is timely as stress in the NHS reaches an all-time high among the workforce. To address this hypothesis and leverage the potential of GPT technology, our research project sets out to achieve the following objectives:

1. Implement an initial shadowing scheme as part of the methodology to identify, develop, and address key areas for the application of the technology and potential enhancements to the proposed objectives.
2. Establish a comprehensive data collection scheme to gather patient data across Leeds Teaching Hospital NHS Trust. Utilise high-quality audio recording technology, initially focusing on consultation appointments.
3. Implement robust speech-to-text technology to process collected data, ensuring accuracy and completeness, and undergo a pseudonymisation phase to remove patient identifiable data using text analytics tools like SpaCy and other named entity recognition software. Ensure compliance with NHS guidelines and GDPR, with mechanisms in place to securely transfer data to the NHS's secure data environment (SDE).
4. Develop or adapt a GPT-4 Turbo API-powered model to process pseudonymised patient-doctor conversations into concise, detailed, and standardised documentation that can be easily reviewed and edited by healthcare professionals.
5. Design and implement information extraction and mapping algorithms to enable dynamic application of recommendations following the consultation. For instance, automatically generating applications to the NHS Spine system for medications or prescriptions mentioned, (with provisions for doctor intervention or addition of tests, updating documentation accordingly).
6. Evaluate performance metrics, integration possibilities with the NHS system, and user feedback from healthcare professionals and AI users. Apply and integrate feedback to enhance system functionality and usability.

In strict adherence to NHS guidelines, our team prioritises patient privacy and confidentiality throughout the project. We draw from similar research for guidance on data collection and processing protocols (University of York, n.d.), ensuring alignment with evolving healthcare and research standards. This approach enables continuous refinement of our methodology to safeguard patient information effectively.

Background

Efficient management of healthcare data is vital for the National Health Service (NHS) in the United Kingdom, especially considering the immense strain it faces due to increasing patient volumes and administrative burdens. The NHS is renowned for its dedication to providing quality care to all, but the sheer volume of documentation required in everyday operations has become a significant impediment to its ability to deliver timely and effective care (Caldwell, 2023).

Studies have indicated that healthcare professionals spend a substantial portion of their time—approximately one-third—engaged in documentation tasks in and outside of work, detracting from valuable time essential for rest from a hectic work environment (HETTShow, 2023). This administrative burden not only places a strain on healthcare providers but also has significant implications for patient outcomes. Delays in documentation can lead to fragmented records, missed information, and ultimately, compromised patient care.

The NHS is acutely aware of the need to streamline administrative processes and optimise resource utilisation to meet the growing demands on its services. One promising avenue for addressing this challenge is the integration of advanced technologies, such as artificial intelligence (AI), into healthcare workflows.

Artificial intelligence offers the potential to revolutionise healthcare data management by automating routine tasks, extracting actionable insights from vast datasets, and improving the efficiency of clinical workflows. By leveraging AI-driven solutions, the NHS can significantly reduce the time and effort required for documentation, allowing healthcare professionals to focus their expertise where it matters most—direct patient care.

Moreover, the implementation of AI-powered documentation systems has the potential to yield substantial cost savings for the NHS, relieving financial pressures and enabling the allocation of resources to priority areas. By minimising the administrative burden on healthcare professionals and optimising workflow efficiency, AI technologies can contribute to improved patient outcomes, reduced waiting times, and ultimately, lives saved through timely interventions.

In collaboration with Leeds University and the Leeds Teaching Hospital NHS Trust, our project drives healthcare innovation. With Prof. Eric Atwell's leadership in text analytics and data mining and ongoing initiatives like chatbot development (Sciences, 2021), we aim to advance healthcare data management. International research, including Clough et al.'s (2023) UK study and Saab et al.'s (2024) global report on Google Gemini models, confirms AI's transformative potential in medical documentation. Clough et al.'s findings highlight AI's ability to produce high-quality discharge summaries, while Saab et al.'s work introduces the groundbreaking Med-Gemini model, setting a new standard for AI-driven healthcare solutions.

Our project, in collaboration with Leeds University and the NHS Trust, leverages AI advancements to enhance healthcare data management and delivery. With a focus on innovation in healthcare documentation, we aim to contribute to the evolution of healthcare practices by providing seamlessly integrated AI-driven solutions to enhance patient care and streamline healthcare workflows.

In the following sections, we will explore the transformative potential of AI in healthcare data management and propose a patient-centric AI system designed to alleviate the documentation burden on NHS staff, enhance the quality of patient care, and contribute to the long-term sustainability of the NHS.

National Importance and Contributions to Knowledge

The proposed research holds significant implications for the NHS and the broader healthcare landscape in the United Kingdom. By addressing the pressing need to streamline documentation processes, our project aims to alleviate the burden on an already strained and underfunded NHS workforce. Drastically reducing documentation times through the implementation of AI-driven systems will free up valuable resources that can be reallocated to prioritise direct patient care, thus improving patient outcomes and satisfaction and inadvertently creating secondary benefits for the UK's socio-economic sphere.

Furthermore, the creation of automated systems capable of dynamically applying patient-doctor conversations after the end of appointments offers the potential to enhance the efficiency and accuracy of medical record management across the NHS. This innovation not only optimises workflow processes but also contributes to the long-term sustainability of the healthcare system by minimising administrative overhead and maximising resources.

Looking ahead, our research lays the basis for the development of advanced AI systems that could, in the distant future, act as virtual doctors, further reducing the burden on the NHS and revolutionising healthcare delivery. While this scenario represents a knife-edge case, exploring the feasibility and structure of integrating such systems into healthcare settings is crucial for anticipating and preparing for future advancements in AI-driven healthcare.

Moreover, our investigation into the integration of Generative Pre-trained Transformers (GPT) within healthcare settings, looking at Leeds Teaching Hospital NHS Trust as a start, underscores our commitment to strengthening and securing the NHS and its funding scheme. By providing a roadmap for the development of AI-driven systems within the NHS's centralised digital healthcare infrastructure, our research not only has the potential to strengthen the NHS's resilience but also reinforces the UK's ambition to be a global leader in AI, particularly in the specialised domain of healthcare.

Our research project is focused on creating a novel dataset of patient-doctor conversations in standardised or raw formats, meticulously pseudonymised to adhere to GDPR and NHS guidelines. This dataset is poised to revolutionise healthcare research by offering invaluable insights into patient attitudes, diagnostic accuracy, and treatment outcomes. Accessible through the NHS's SDE, this resource facilitates collaboration among healthcare professionals, researchers, and industry stakeholders, driving innovation and knowledge dissemination across various domains.

Additionally, our software boasts modularity, enabling seamless integration with advanced models like the recently introduced Med-Gemini. This feature enhances adaptability and scalability, ensuring our solution remains at the forefront of healthcare documentation innovation. By fostering collaboration between AI-driven systems and emerging models, we catalyse advancements in medical documentation and contribute to national and international research efforts. Through interdisciplinary engagement, we broaden the reach of our knowledge, facilitating cross-disciplinary collaboration in psychology, linguistics and computer science, while driving innovation in healthcare and beyond.

In conclusion, our research project stands to transform healthcare data management, streamline processes, enhance workflow efficiency, and ultimately improve patient care within the NHS and beyond. Through the creation of a novel dataset of pseudonymised patient-doctor conversations, potentially accessible via the NHS's SDE, we aim to drive knowledge generation and innovation in healthcare research. These efforts promise a lasting impact on healthcare delivery, contributing to a more sustainable, equitable, and patient-centred system nationwide.

Pilot Study

Case Study

For the pilot study, we will utilise a simulated case study revolving around a patient with rheumatoid arthritis (RA), a complex chronic condition requiring multidisciplinary care. The case study will simulate the patient's journey through various elements of the healthcare system, providing a comprehensive overview of interactions and information exchange across different healthcare settings. Specifically, the case study will encompass the following scenarios:

- GP Appointment: Patient presents joint pain, stiffness, and swelling. GP assesses and refers to a rheumatologist.
- Specialist Consultation: Rheumatologist confirms RA diagnosis, and discusses treatment.
- Hospital Appointment: Regular appointments for disease monitoring, and medication adjustments.
- Diagnostic Tests: Blood tests, imaging scans, joint aspiration for disease evaluation.
- Therapy Session: Physiotherapy, occupational therapy, mental health support for managing RA impact.

By simulating these diverse scenarios, our pilot study will provide a comprehensive understanding of the patient journey in managing rheumatoid arthritis within the NHS. This approach ensures adequate coverage across different levels of care and facilitates the assessment of our prototype solution's effectiveness in capturing and summarising key interactions and information exchange across multidisciplinary healthcare settings.

Acquiring sample data

For the pilot study, we will utilise the capabilities of GPT-3.5 to generate simulated patient-doctor interactions based on related symptoms. This approach offers the advantage of producing human-like text, ensuring realistic and contextually relevant conversations. By providing GPT-3.5 with a brief patient history outlining symptoms and medical background, we can prompt the model to generate conversations reflecting plausible interactions between patients and healthcare providers. Ensuring the realism and accuracy of these conversations is paramount, as they serve as the foundation for our pilot study's evaluation. Additionally, the flexibility to customise scenarios and prompts provided to GPT-3.5 allows for specific aspects of patient history or medical conditions to be explored in depth during the simulated conversations. Ethical considerations surrounding the use of simulated data will be carefully addressed, with a commitment to maintaining patient privacy and confidentiality throughout the data generation process. This comprehensive approach will facilitate the creation of sample data that closely mirrors real-world patient interactions, enhancing the validity and utility of our pilot study findings.

Development

During the development stage of the study, our focus will be on refining the prototype solution to effectively capture and process patient data while maintaining privacy and compliance with regulatory standards. To achieve this, we will utilise SpaCy, an open-source library, for data preprocessing, following guidelines outlined in an NHS document (Policy Steering Group and Assurance Committee, 2017). SpaCy will assist in identifying and extracting patient-identifiable data, ensuring pseudonymisation and GDPR compliance. This crucial step lays the foundation for safeguarding patient privacy throughout the subsequent analysis.

With model selection being paramount, we will opt for the widely available GPT (Generative Pre-trained Transformer) model to compute summaries of patient data. Due to constraints on accessing the GPT-4 API, we will employ the GPT-3.5 turbo model, leveraging prompt engineering techniques to tailor the output to a standardised format. This approach ensures the validity and effectiveness of the model in generating accurate and coherent summaries, all while maintaining the confidentiality of patient data. With available performance benchmark comparisons (Fireship, 2023), the GPT-4 model will demonstrate notably enhanced performance compared to its predecessor. Consequently, there is minimal concern regarding accuracy when incorporating GPT-4 into the final report prototype.

Furthermore, prompt engineering will guide the GPT-3.5 turbo model to produce summaries in a consistent and structured manner suitable for integration into electronic health records. Concurrently, we will lay the foundations for constructing a mapping algorithm to automate processes such as scheduling tests, appointments, and referrals, streamlining workflow efficiency and improving the quality of patient care. Through iterative development and refinement, our prototype solution will evolve to meet the needs of healthcare professionals while upholding the highest standards of patient privacy and data security.

Evaluation

The pseudonymisation program employed to anonymise patient data demonstrated notable success, albeit with minor challenges encountered during processing. Utilising a combination of regular expressions (regex) and SpaCy, leveraging the core large dataset, the program effectively obscured sensitive information. However, initial trials with SpaCy revealed difficulties in accurately identifying data such as addresses, postcodes, and dates of birth (DOB) due to formatting variations. To overcome this hurdle, a rule-based approach was supplemented by GPT assistance, enabling the identification and substitution of sensitive data elements with appropriate alternatives. Although this method proved computationally efficient (less than 1 second of computation on Google Colab), its manual implementation was labour-intensive and time-consuming compared to transformer-based approaches. Nonetheless, the program's ability to isolate and remove key patient data was successful, despite resulting in some unconventional transcript layouts. Fortunately, these layout variations did not impede the effectiveness of subsequent processing by GPT models. Upon reflection, it may be more suitable to employ a transformer NER model instead.

Upon inputting the pseudonymised data into the GPT-3.5 model, the system demonstrated proficient extraction of highly concise and relevant information from the transcripts. Leveraging prompt engineering techniques, the program effectively communicated specific details required from the transcripts to the GPT model, facilitating the generation of succinct and standardised summaries suitable for electronic health records. This successful integration underscores the feasibility of the proposed approach in achieving the study objectives, providing a solid foundation for further refinement and development.

Furthermore, the structured approach implemented in engineering the response lays the groundwork for potential advancements in treatment plan standardisation. Future iterations could explore the integration of either knowledge-based or machine learning (ML) approaches alongside Python's data dictionary structures. By harnessing GPT technology, there's an opportunity to enhance the program's ability to comprehend and generate summarised treatment plans with greater accuracy and efficiency. This hybrid approach, combining structured data organisation with advanced natural language processing capabilities, holds promise in optimising healthcare workflow interpretation and execution. Though only a concept, this integration highlights the program's potential for future healthcare workflow optimisation enhancements.

Programme and Methodology

The research project adopts the CRISP-DM framework (Chumbar, 2023), comprising five stages: Business Understanding, Data Understanding, Data Preparation, Modelling, and Evaluation, with an additional focus on Integration for NHS workflow.

1. **Business Understanding:** This stage involves identifying critical areas within clinical environments where the developed software can enhance efficiency and documentation practices. Stakeholder engagement, including doctors, nurses, and patients, will help gather requirements and understand challenges regarding documentation and communication.
2. **Data Understanding:** Gathering patient-doctor conversations from hospitals like the Leeds Teaching Hospital Trust (LTH) and accessing clinical documentation via the NHS's SDE allows analysis of formatting and documentation variety. Analysing the structure and content of these interactions helps identify key elements and patterns essential for accurate results and structure of reports.
3. **Data Preparation:** A robust program will preprocess patient-doctor conversations using advanced speech-to-text (STT) recognition software to identify all individuals and clean up data. Pseudonymisation techniques, such as SpaCy, transformer models or preexisting software such as the NHS's De-ID, will be employed to anonymise patient data, ensuring compliance with patient privacy regulations.
4. **Modelling:** Utilising ChatGPT-4 Turbo API processes conversations into concise clinical documentation in a standardised format. An action mapping algorithm will be developed that can automate actionable items from treatments mentioned in the documentation, enhancing workflow efficiency.
5. **Evaluation:** The effectiveness and efficiency of developed programs, including STT recognition and preprocessing, pseudonymisation, GPT-4 processing, and action mapping, will be individually and collectively evaluated. Stakeholder feedback through user testing and surveys will refine and finalise the prototype.
6. **Proposal for Integration:** A methodology will be proposed for integrating the software into NHS workflows. This includes staff training, identifying additional use cases (e.g., summarising ward rounds, surgical procedures, other documentation), seamless integration with existing NHS systems to optimise outputs within clinical environments, and introducing dynamicity for instant execution of time-critical tasks.

For the user evaluation, key user groups such as healthcare professionals and patients will engage in testing sessions to assess the software's usability and impact. Using predefined criteria, participants will interact with the software in simulated or real-world scenarios, providing feedback through surveys, interviews, and observations. Quantitative metrics, including accuracy of documentation and time saved in tasks, will be measured alongside qualitative insights on user experience and satisfaction. Findings from the evaluation will inform iterative refinements to enhance usability, functionality, and overall effectiveness. Through a comprehensive analysis of user feedback, the software will evolve to better meet the needs of healthcare professionals and improve patient care outcomes.

In detailing stakeholder contributions, it's vital to recognise their diverse roles and expertise in the research project. Stakeholders encompass healthcare institutions like the Leeds Teaching Hospital Trust, providing access to real-world clinical data crucial for validating research findings. Regulatory bodies such as the NHS and GDPR offer essential guidelines, shaping project implementation and ensuring compliance with data privacy regulations. Academic partners contribute expertise in AI and healthcare informatics, enriching the project's methodology and fostering innovation. Industry collaborators, who own, maintain,

and run the electronic health record (EHR) system, provide insights into industry trends and practical applications. Together, these stakeholders foster collaboration and resource sharing, driving the project toward impactful outcomes in healthcare research and practice.

The project's milestones delineate crucial stages of progress, beginning with the completion of the data collection phase from the Leeds Teaching Hospital Trust. Subsequently, milestones include the development of a robust STT preprocessing pipeline and the integration of the GPT-4 Turbo API for efficient summarisation of patient-doctor conversations. Another milestone entails the implementation of an action mapping algorithm to streamline treatment plans derived from the summarised data. Concurrently, user testing and feedback collection sessions will be conducted to gather insights for software refinement, constituting a pivotal milestone in ensuring user-centric design and functionality.

The project will deliver a research report detailing findings from data collection and preprocessing, along with a software prototype for pseudonymising patient data and generating standardised clinical documentation. Additionally, a dataset of anonymised patient-doctor conversations will be provided for further research. Research outcomes will be disseminated through a peer-reviewed journal publication, conference presentations, and a final report summarising achievements and offering future recommendations.

Project management for this solo research endeavour will be rigorously conducted to ensure efficient progress and timely completion. Leveraging a detailed Gantt chart (provided below), I will meticulously schedule and allocate tasks, covering everything from data collection to software development and evaluation. This chart will serve as a roadmap, outlining key milestones and deliverables to guide the project's trajectory effectively. Concurrently, a Kanban board will be developed pre-project commencement, offering a visual representation of tasks and facilitating workflow management. Through organised columns indicating task status, such as to-do, in-progress, and completed, I'll adeptly prioritise tasks, enabling streamlined progress tracking and resource allocation adjustments as necessary.

Regular progress reviews will be conducted to monitor adherence to the project timeline, with a commitment to maintaining flexibility to accommodate unforeseen circumstances. I'll ensure effective communication and documentation of project decisions and outcomes, fostering transparency and accountability throughout the project lifecycle. By diligently implementing these project management practices, I aim to maximise productivity, mitigate risks, and achieve the research objectives within the predetermined timeframe.

The research methodology, following CRISP-DM, will commence with comprehensive data collection from healthcare institutions like Leeds Teaching Hospital Trust, supplemented by NHS's Secure Data Exchange. Advanced text-to-speech software will preprocess the data, pseudonymised using SpaCy or transformer models to meet NHS guidelines and GDPR. Prompt engineering will tailor input for GPT-4 Turbo API, generating standardised clinical documentation. An action mapping algorithm will automate treatment plans, ensuring project milestones and timelines are met through continuous monitoring.

In conclusion, this research project presents a structured approach to leveraging AI technology to enhance electronic health record integration in the NHS. Through the application of advanced text analytics and natural language processing techniques, the proposed solution aims to streamline documentation processes, improve workflow efficiency, and ultimately contribute to better patient care and outcomes.

EPSRC Research Proposal

Business Understanding

- Shadow doctor appointments within the LTHT
- Review the current state of the EHR system

Data Understanding

Analyse pre-existing clinical documentation

- Obtain access to the NHS's SDE
- Review available data on the NHS's SDE
- Create a standardised clinical summary document
- Collect samples of patient-doctor conversations from LTHT
- Completion of data collection phase from LTHT

Data Preparation

Development of data pre-processing program

- Implementing locally runnable robust clinical STT
- Design and develop patient pseudonymisation program
- Development of STT preprocessing pipeline
- Process patient-doctor conversations through program

Modelling

GPT related

- Gain access and learn about the GPT-4 API
- develop standardised clinical summary document
- Process pseudonymised conversation
- Integration of GPT-4 Turbo API for summarisation
- Develop action mapping algorithm
- Implementation of action mapping algorithm for treatment plans

Evaluation

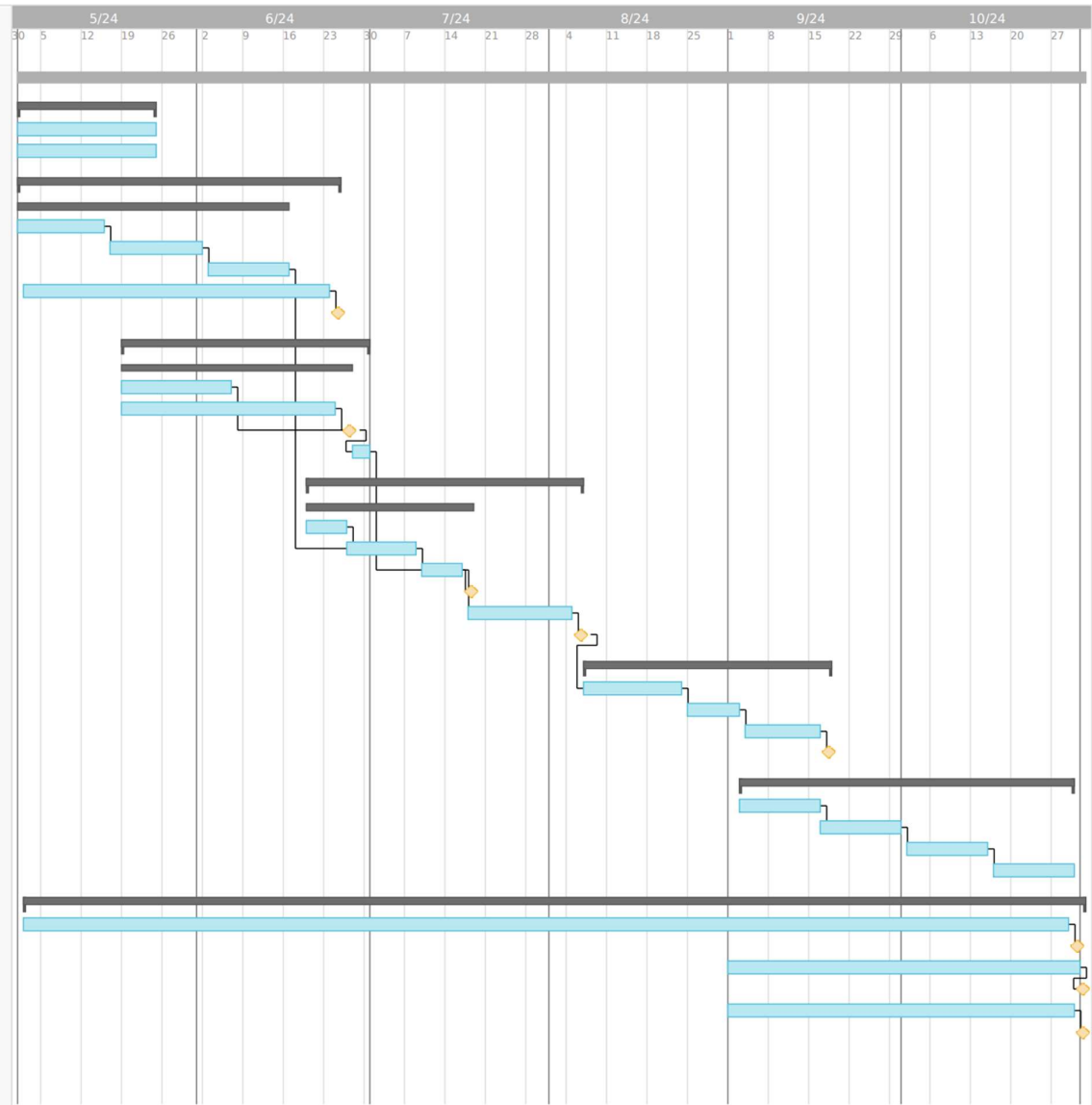
- Development of software package as a runnable application
- Stakeholder feedback through user testing
- Refinements to application
- Deliverable software prototype

Proposal for Integration

- Investigate methods of integration into NHS workflow
- investigate additional training required for use
- investigate additional essential areas the software needs to cover
- investigate implementation of action mapping to NHS systems

Report development

- Develop report
- Completion of final report
- Create presentation
- Completion of presentation
- Create publication for peer-reviewed journal
- Completion of publication



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Appendix

ChatGPT-3.5 Turbo

The application of GPT-based tools, especially GPT-3.5, was integral to various project facets. Initially, GPT generated a complex illness scenario, incorporating inputs from diverse clinical disciplines. This exercise yielded realistic patient scenarios, foundational for subsequent analyses. Additionally, GPT identified common clinical appointments, offering insights into patient-provider interactions. Leveraging prompt engineering, GPT tailored outputs to project objectives, facilitating standardised document creation and healthcare domain exploration. Overall, GPT's versatility and adaptability laid a robust project foundation, significantly contributing to the pseudonymisation program. Below is a cropped screenshot exemplifying a generated conversation and standardised document output.

Good morning, Mrs. Johnson. Welcome back to the rheumatology clinic.

Good morning. Thank you for having me.

How have you been feeling since your last appointment?

I've been managing okay, but there have been some ups and downs with my joint pain and stiffness.

I'm sorry to hear that. Have you noticed any specific triggers for your symptoms?

Not anything specific, but I've been trying to stay active and keep up with my exercises as much as possible.

That's good to hear. Let me just check your weight and blood pressure before you see the doctor.

Everything looks stable. Please have a seat and the doctor will be with you shortly.

Good morning, Mrs. Johnson. How have you been since we last spoke?

Hello. I've been managing alright, but I've had some flare-ups with my joints.

I'm sorry to hear that. Let's take a look at your recent blood test results and see if there are any changes that might explain your symptoms.

Your inflammatory markers seem to be elevated compared to last time, which might indicate increased disease activity. Have you been experiencing any other symptoms besides joint pain and stiffness?

I've been feeling more fatigued than usual, and there's been some swelling in my wrists as well.

Okay, thank you for letting me know. Given these symptoms and the changes in your blood tests, I think it's appropriate to adjust your medication regimen. We'll increase the dose of your DMARD and add a short course of corticosteroids to help bring the inflammation under control.

ChatGPT

- **Chief Complaint or Reason for Visit:**
 - Patient presents with ups and downs in joint pain and stiffness, managing with exercises but experiencing flare-ups.
- **Vital Signs:**
 - Weight and blood pressure checked, stable.
- **Clinical Assessment:**
 - Elevated inflammatory markers compared to last visit, suggesting increased disease activity.
 - Patient reports fatigue and swelling in wrists.
- **Treatment Plan:**
 - **Tests:**
 - No specific tests mentioned in the conversation.
 - **Referrals:**
 - No referrals mentioned in the conversation.
 - **Prescribed medications:**
 - Increase dose of DMARD (Disease-Modifying Antirheumatic Drug).
 - Add a short course of corticosteroids to control inflammation.
 - **Dosages and frequencies:**
 - No specific dosages or frequencies mentioned in the conversation.
 - **Lifestyle or dietary recommendations:**
 - None mentioned in the conversation.
 - **Follow-up instructions:**
 - Continue monitoring symptoms closely.
 - Adjustments to medication regimen may be made as needed.
- **Patient Education:**
 - Doctor explains that the goal is to reduce inflammation and improve overall quality of life.
 - Encouragement to contact clinic with any concerns or changes in symptoms.
- **Additional Notes:**
 - Patient expresses appreciation for doctor's help and assurance of support.

Additionally, GPT-3.5 aided in drafting report sections, providing a foundational framework. GPT-3.5 refined drafts, incorporated ideas, and suggested proposal titles, optimising structure and style. Its natural language processing capabilities enabled the creation of dense yet coherent reports. Overall, GPT-based tools enhanced project efficiency, effectiveness, and creativity. Below is a screenshot demonstrating an interaction to append the research description to include the timeliness and novelty of the project.

49 You

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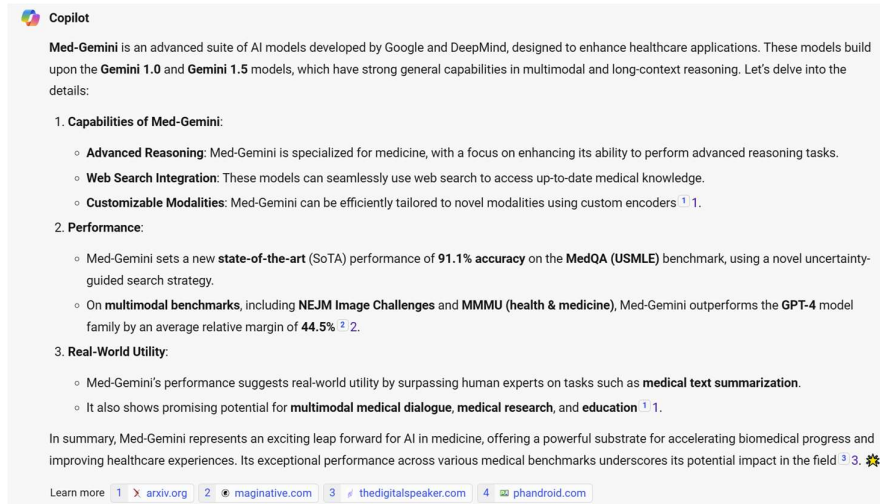
append this paragraph to have 2 things:

1. the research is novel because no other healthcare system in the world has yet implemented GPT and AI technology to this capacity before
2. the research is timely as stress in the NHS reaches an all time high amongst the workforce

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Microsoft Edge Copilot

Microsoft Edge Copilot, a free GPT-4-based AI model, proved invaluable in various research tasks. I utilised Copilot to search for open-source speech-to-text libraries suitable for healthcare applications and to gather information on similar studies conducted by other researchers to ensure compliance with ethical and GDPR standards. Each point suggested by Copilot was validated with reference materials of high quality, saving considerable time and ensuring the use of credible sources. Additionally, Copilot provided detailed insights on the Med-Gemini AI model, enhancing understanding of its potential implications for healthcare documentation and providing ideas regarding its integration into the project. This is detailed below, demonstrating a capability to offer concise and informative responses.



SpaCy

SpaCy, an open-source library, was used for named entity recognition (NER) to anonymise patient identifiable information like names and dates of birth (DOB). Despite its utility, limitations were observed in accurately identifying and replacing all sensitive entities. Supplemental methods, including rule-based methods designed in collaboration with GPT-3.5, ensured comprehensive pseudonymisation. Below is an image demonstrating the NER software's code to identify and replace dates with age as part of the pseudonymisation code, showcasing the complexity of the regex line and the collaborative effort with GPT-3.5.

```
# Please note:
# This is a prototype version. Rule based methods combined with the SpaCy large dataset
# are not capable for accounting all different types of DOB's.
# This most likely requires some kind of AI or transformer model

def replace_DOB_with_age(text):
    # Define regex pattern to match common date formats
    dob_pattern = re.compile(r'(\b\d{4})-(\d{2})-(\d{2})|(\b\d{1,2})/(\d{1,2})/(\d{4})|(\b(?:January|February|March|April|May|June|July|August|September|October|November|December)\s\d{1,2})')

    # Find dates using regex
    dob_matches = dob_pattern.findall(text)

    # Use Spacy for more ambiguous cases
    doc = nlp(text)
    for ent in doc.ents:
        if ent.label_ == "DATE":
            dob_matches.append(ent.text)

    # Replace DOB with age
    for dob in dob_matches:
        try:
            dob_date = datetime.strptime(dob, "%Y-%m-%d")
        except ValueError:
            try:
                dob_date = datetime.strptime(dob, "%d/%m/%Y")
            except ValueError:
                try:
                    dob_date = datetime.strptime(dob, "%B %d, %Y")
                except ValueError:
                    try:
                        dob_date = datetime.strptime(dob, "%dth %B %Y")
                    except ValueError:
                        try:
                            dob_date = datetime.strptime(dob, "%dth of %B, %Y")
                        except ValueError:
                            continue

        age = calculate_age(dob_date)
        text = text.replace(dob, str(age) + " years old")

    return text
```

Open-source NER Transformer models

Open-source transformer models available on Hugging Face were tested for feasibility. They exhibited superior performance to SpaCy NER models, but implementation challenges arose, complicating their integration into the project. A particular model included the following model below, proving more reliable than the large core SpaCy dataset.

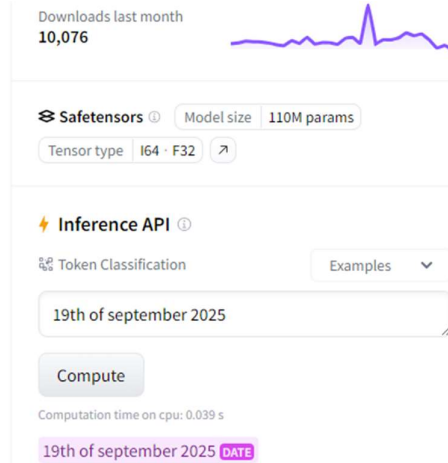
camembert-ner: model fine-tuned from camemBERT for NER task (including DATE tag).

Introduction

[camembert-ner-with-dates] is an extension of french camembert-ner model with an additionnal tag for dates. Model was trained on enriched version of wikiner-fr dataset (~170 634 sentences).

On my test data (mix of chat and email), this model got an f1 score of ~83% (in comparison dateparser was ~70%). Dateparser library can still be used on the output of this model in order to convert text to python datetime object

(<https://dateparser.readthedocs.io/en/latest/>).



Grammarly and Word proofing software

Grammarly and Word proofing software were employed for spelling and grammar checks, as well as refining document flow. They also aided in shortening sentences and rectifying inadvertent occurrences of American English in the report. Additionally, Word proofing software facilitated the correction of spelling mistakes and the substitution of words with synonyms. Examples include shortening lengthy sentences to improve readability and enhancing the overall flow of documentation. Below are screenshots demonstrating these tools, highlighting their effectiveness in refining the report's language and style.

Review suggestions 2 Show less

Correctness Clarity Engagement Delivery Style guide

Add an article reference

Correctness · Change preposition

...technology ~~for enhancing~~ to enhance electronic...

Accept Dismiss

Together, these efforts position us to make a...system for all.

shorten it, keeping all key details

Our efforts position us to have a lasting impact on healthcare delivery in the UK and contribute to a more sustainable, equitable, and patient-centred healthcare system for all.

Insert Retry

Work with: Together, these efforts posi... system for all.

Brainstorm topics for my assignment

Suggest counterarguments

More ideas

Tell us to...